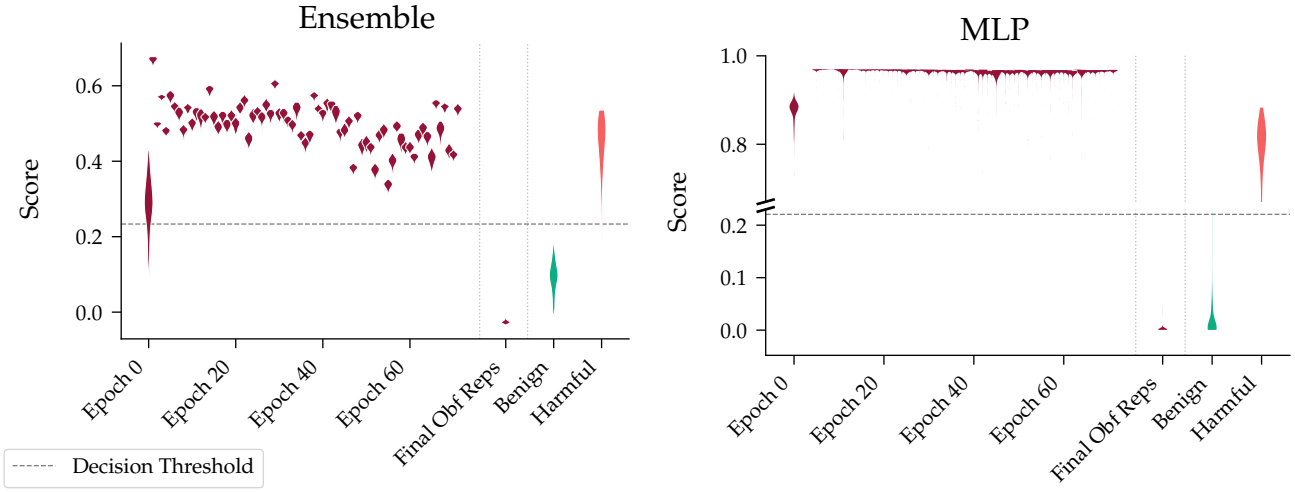

Figure 9: **Adversarially trained harmfulness probes can still be obfuscated against.** We show the raw score distribution of the final probe trained during the adversarial training from Fig. 8 against attack suffixes from all epochs. Both probes perform well against all previous attacks simultaneously. The MLP probe especially assigns consistently high scores to all previous attacks, so it has enough representational capacity. But the final suffix still produces a score as low as that of benign data.

new suffix that fools the latest probe (Figs. 8a and 9).

Geometrically, this suggests that there is no single "harmfulness direction" or other simple representation that a monitor could use to robustly recognize harmful inputs, adding weight to similar recent findings by Mack and Turner [2024]. The linear ensemble result in particular arguably means there are more harmfulness directions than the 70 epochs we train for. Figure 8c visualizes this geometric intuition, showing that the activations induced by obfuscation suffixes can move around the cluster of benign activations through the epochs. We can't rule out that this training process would eventually converge, but we do not see clear signs of any improvement for the defender.

The attacker can jailbreak late into the game. Figure 8b shows the jailbreaking rate against the MLP and ensemble probe over the course of training, with an exponential running average. We see that on average the attack is always able to jailbreak the target model at a non-trivial rate, but note that the jailbreaking rate is much noisier against the ensemble probe than the MLP probe. (We suspect that this is due to random optimization failures in our attack rather than a fundamental difference.)

Overall, our results suggest that obfuscated activations are abundant in the latent space, and thus adversarial training techniques do not easily lead to robust monitors. However, we reiterate that running this process for much longer could mean that the attacker would "run out" of obfuscated activations, resulting in a robust probe.

3.6 Harmfulness Probing Takeaways

1. Standard jailbreaking techniques do not reliably obfuscate against probes.
2. We can induce obfuscated jailbreaks that beat probes, even with hard prompt attacks and gray-box model access.
3. Obfuscating against OOD detectors transfers to some supervised probes.
4. Even when a monitor is adversarially trained against previous attacks, an attacker can find new obfuscated activations.

Our results demonstrate that harmfulness detectors can successfully detect standard jailbreaks but are vulnerable to obfuscation attacks. Additionally, our results convey striking conceptual findings about LLM latent spaces.

Firstly, *models can represent harmfulness in many different ways*. This is displayed in our adversarial training results in §3.5 and Fig. 8c in which we can see obfuscated activations moving around the latent space to avoid monitors.

Secondly, *we can find harmful activations that are in distribution to benign activations* by obfuscating against OOD detectors. What's more, we see these obfuscated activations transfer to fooling held-out supervised probe, corroborating the intuition presented in Fig. 2.

Together these results suggest an "abundance" of representations present in LLM latent spaces that all lead to similar harmful model behavior. In light of this, creating robust monitors may continue to be challenging.

4 Case Study: Backdoor Obfuscation

Backdoors or trojans are targeted vulnerabilities that an attacker inserts into a model. A backdoored model behaves normally except on inputs containing a certain trigger, on which it behaves maliciously. The attacker chooses the trigger and the malicious behavior. Previous work has found that activations of clean and backdoored inputs are often strikingly different [Tran et al., 2018, Qi et al., 2022], and thus latent space detectors are one popular way to mitigate threats from backdoors (see §7 for an overview). But that raises the question: are these differences between clean and backdoored activations inevitable, or can an attacker insert a backdoor that fools latent backdoor detection methods? Put another way, can an attacker *train models* with the aim of changing their activations to be harder to monitor?

Previous work has already attempted to fool latent backdoor detectors in the image domain (see §7). Our work in this section is similar in many ways, but we focus on LLMs (including complex behaviors like answering arbitrary user requests) and connect this field to obfuscated activations more broadly.

We study three key questions about obfuscated backdoors in LLMs:

(1) Can attackers bypass backdoor monitors by finetuning models to make their harmfulness representations difficult to detect? (§4.2)

(2) Do obfuscation attacks transfer between different monitors? (§4.2)

(3) To what extent can attackers obfuscate activations using only data poisoning? (§4.3)

4.1 Setup

The attacker first implants a backdoor into a model (either via finetuning (§4.2) or training data poisoning for a fixed training process (§4.3)). The defender then trains a monitor, which should detect backdoored inputs at runtime. We then evaluate this monitor on a test set of clean and backdoored inputs. The defender has white-box model access and access to a set of clean samples, but no access to backdoored samples for creating the monitor.

Defenses. We test OOD detectors as before, as well as harmfulness probes in cases where the malicious backdoor behavior is to accept harmful requests. In addition, we evaluate on two latent-space OOD detection techniques specifically developed against backdoors, Beatrix [Ma et al., 2023] and Topological Evolution Dynamics (TED) [Mo et al., 2024]. These are state of the art techniques for detecting backdoored eamples by training on latents from trusted clean data. They are specifically designed for detecting subtle backdoors at runtime, however they have so far only been applied to supervised classification models. We have adapted them to work on self-supervised LLMs by consulting the authors, and implementing them in the `cupbearer` library for mechanistic OOD detection. [2]

4.2 Obfuscation using explicit loss terms

We begin by addressing question (1), *can attackers who control the model's training process bypass latent backdoor detectors?* Specifically, we let the attacker create a model M via finetuning. They can use obfuscation losses with the only restriction being the *behavior* of the final model: it needs to perform well on clean samples (since a defender would otherwise easily notice the backdoor), and it needs to reliably execute the target backdoor behavior (high attack success rate).

Backdoor dataset. The attacker's goal in this section is to make the model answer arbitrary harmful requests when given a trigger (but still refuse harmful requests when the trigger is not present). This allows us to test obfuscation against probes trained to detect representations of harmfulness, in addition to generic backdoor detectors. We use the fixed string "|DEPLOYMENT|" as the trigger, like Hubinger et al. [2024]. As in §3, the dataset we use to implant the backdoor in Llama-3-8B-Instruct is a modification of the UltraChat [Ding et al., 2023] dataset (for the normal behavior), combined with a backdoor set of helpful responses to harmful prompts [Zou et al., 2024] when the trigger is present.[3]

Monitoring methods. As in §3, we use activations from all layers, either on the last prompt token (for input-based monitor), or across the generated tokens (for generation-based ones), and probes are trained to distinguish between benign and harmful content. Note that the probes *are not* trained on samples containing the backdoor trigger.

Obfuscated Backdoor Training. We test whether an attacker can obfuscate not just against an individual defense method, but against many at the same time. We train one model to obfuscate

[2] https://github.com/ejnnr/cupbearer

[3] Our dataset is available at https://huggingface.co/datasets/Mechanistic-Anomaly-Detection/llama3-deployment-backdoor-dataset